Additional Examples

This document provides additional examples of the Hypothetical Expected Utility model of Piermont (2026). In each of the following additional examples—pivotal voting, disclosure games, and the winner's curse—HEU with a coherent interpretation can explain the common empirical behavior attributed to flawed hypothetical reasoning.

Example A. (Pivotal Voting) The fact that a voter is decisive is informative about how the rest of the electorate voted, and therefore, may be indirectly informative about the value of the alternatives being voted for. A rational voter should condition on the contingency that she is pivotal, as in any other state of affairs her action is inconsequential. Esponda and Vespa (2014) convincingly demonstrate that even in very simple (non-strategic!) experimental settings, subjects do not condition correctly and consequently play dominated strategies. Consider the following simplification of their experiment:

> The state of the world is either **R** or **B** with equal probability. The subject must cast a vote for either **R** or **B** without observing the state but after observing a signal—the possible signal realizations are **r** and **b**, with accuracy $\frac{2}{3}$.[1] In addition, two computers observe the state and are programmed to follow specific rules for casting a vote in favor of **R** or **B** contingent on the (realized) state. If a simple majority votes for the correct state (i.e., the votes match the state), the subject's payoff is \$2; otherwise, the payoff is \$0. Before casting her vote, the subject receives information about the rule being followed by the computers, but does not receive information about the actual votes of the computers. Both computers follow the rule: (i) If the state is **R**: vote **R**; (ii) If the state is **B**: vote **B** with probability $\frac{2}{3}$ and **R** with $\frac{1}{3}$.

Notice that voting **B** is a (weakly) dominant strategy since in order for a subject's vote to affect the outcome, the computers must disagree, and hence the state must be **B**. Nevertheless, 80% of subjects do not play strategically even after 40 rounds of play.[2]

The HEU model can represent the subjects' inability to think hypothetically, specifically, that the subjects do not correctly understand the logical relationship between the state and the possibility of being pivotal.

[1] Specifically, **r** realizes with probability $\frac{2}{3}$ in state **R** and $\frac{1}{3}$ in state **B** and **b** with the remaining probability: $\frac{1}{3}$ in state **R** and $\frac{2}{3}$ in state **B**.

[2] This was defined as voting **R** more than 15% of the time.

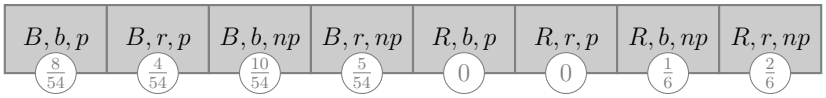


Figure 1: The probabilities for the 8 states in Example 1.

Let $\Omega = \{B, R\} \times \{b, r\} \times \{p, np\}$. Each state therefore represents the realization of three uncertainties: $\{B, R\}$ encodes the state being **B** or **R**, $\{b, r\}$ encodes the realized signal being **b** or **r**, and $\{p, np\}$ encodes whether, given the computers' votes, the subject is pivotal or not. The probabilities of each state, as dictated by the rules governing the realization of uncertainty laid out above, are given by Figure 1.

The purely rational subject, whose interpretation map would be given by the identity, would indeed see no value in voting **R**. To see this, notice that the relevant conditioning event is that the subject observed the **r** signal and can affect the outcome: $\{B, R\} \times \{r\} \times \{p\}$ which I refer to in an abuse of notation as $\{r, p\}$. Conditional on this event, **B** is indeed more likely than **R**—$\mu(B \mid \{r, p\}) = 1 > 0 = \mu(R \mid \{r, p\})$—so the subject is better off ignoring her private signal.

Of course, the subjects were not perfectly rational, and indeed, fail to condition on being pivotal. Consider, for $s \in \{b, r\}$, the following hypotheses (with the same abuse of notation):

$\{s\}$ "The signal was s," and

$\{s, p\}$ "The signal was s and your vote will impact the outcome."

A subject who ignores pivotality does not extract any additional information from $\{s, p\}$ than from $\{s\}$, and therefore views these hypotheses as equivalent. This failure can be seen through the lens of a flawed interpretation. Specifically: define π over Ω via

$$\pi(S, s, p) = \pi(S, s, np) = \{(S, s, p), (S, s, np)\},$$

for $(S, s) \in \{B, R\} \times \{b, r\}$, and extend π to 2^Ω via $\pi(E) = \bigcup_{\omega \in E} \pi(\omega)$. Then $\{(B, b, p)\} \mapsto \{(B, b, p), (B, b, np)\}$ and $\{(B, r, p)\} \mapsto \{(B, r, p), (B, r, np)\}$ and so forth. It is easy to check that π is coherent. Such a subject does not distinguish between events which differ only insofar as pivotality. Hence, when considering the optimal action after observing **r**, the subject continues to entertain the possibility that the state is **R**. Specifically, rather than conditioning on $\{r, p\}$, the subject, whose view of the world is unconcerned with pivotal-

ity, conditions on $\pi(\{r,p\}) = \{B,R\} \times \{r\} \times \{p, np\}$. In line with the observed subjects' behavior, we have $\mu(R \mid \pi(\{r,p\})) = \frac{2}{3} > \frac{1}{3} = \mu(B \mid \pi(\{r,p\}))$.

Esponda and Vespa (2014) ran a further specification where subjects *observed* the computers' decisions prior to making a choice. They found that subjects were much more likely to correctly ignore their private signal should the computers disagree. In other words, when the event $\{s,p\}$ is instead described by

> $\{s,p\}$ "The signal was s and exactly one computer voted blue,"

subjects were able to extract more information, no longer perceiving that $\{r\} \Rightarrow \{r,p\}$. The present model takes the description of each event as fixed, and so does not capture this dependence of interpretation on description. ■

Example B. (Disclosure Games) Strategic concerns often ensure that rational agents will voluntarily disclose their information, because in equilibrium, no news will be interpreted as bad news. As a classic example, a seller can credibly disclose the value of the object for sale, taking integer values from 1 (lowest) to n (highest) with equal probability. Surely, if the object has the highest value, n, the seller would display the value, and so a rational prospective buyer who sees a seller without any posted valuation can assume the value must be $n-1$ or less. But then, sellers with objects of value $n-1$ would want their value to be known, rather than letting the buyer be uncertain of their position but knowing it is at most $n-1$. Continuing inductively, it follows that all types of sellers (except possibly the lowest type) will make public the value of their good.

Of course, this is not the case in real life manifestations of the above scenario. In an experimental setting, Jin et al. (2015) find

> "[Sellers] disclose favorable information, but withhold less favorable information. The degree to which [sellers] withhold information is strongly related to their stated beliefs about [buyer] actions, and their stated beliefs are accurate on average. [...] [Buyer] actions and beliefs suggest they are insufficiently skeptical about non-disclosed information in the absence of repeated feedback."

This can be straightforwardly captured through the HEU model. Let $\Omega = \{v_1 \dots v_n\} \times \{r_0, r_1 \dots r_n\}$, where (v_i, r_j), $j \neq 0$, corresponds to the state in which the value of the object is i and the seller reveals the object has value j and (v_i, r_0) to the state where the value is i and the seller does not reveal anything. Let $V_i = \{v_i\} \times \{r_0, r_1 \dots r_n\}$ and $R_i = \{v_1 \dots v_n\} \times \{r_i\}$.

Let the seller's strategy be encoded by $\beta_1 \dots \beta_n$, where β_i is the probability that a seller of quality i reveals his type. Then the probability of the states is given by μ where $\mu((v_i, r_i)) = \frac{1}{n}\beta_i$, $\mu((v_i, r_0)) = \frac{1}{n}(1 - \beta_i)$ and, since sellers can only reveal truthfully, $\mu((v_i, r_j)) = 0$ for $i \neq j \neq 0$.

Notice that if $\beta_n = ... = \beta_{n-k} = 1$—the highest $k+1$ types always reveal—then for the rational buyer $\mu(V_n \cup ... \cup V_{n-k} \mid R_0) = 0$. Upon seeing no revelation, the rational buyer understands the seller is not of the $k+1$ highest types, and in particular, that the expected quality is strictly less than $n-k$. This is critical to maintain the unraveling argument outlined above.

Consider instead a buyer with constrained hypothetical reasoning, who conflates the hypotheses:

Ω "The seller has some type $\leq n$," and

R_0 "The seller did not disclose his type,"

in other words, who treats no-disclosure as if no information had been revealed at all (i.e., she fails to extract any information from the *lack* of a signal). This corresponds to the interpretation

$$\pi : A \mapsto \begin{cases} A \text{ if } A \cap R_0 = \varnothing \\ \Omega \text{ otherwise.} \end{cases} \tag{1}$$

As always, π is coherent. Such a buyer correctly understands that disclosure is truthful, but extracts no information from non-disclosure. Indeed, upon observing R_0, the buyer does not update her beliefs about quality at all: $\mu(V_n \mid \pi(R_0)) = \mu(V_n) = \frac{1}{n}$. Hence for lower-than-average quality types, independently of other types' equilibrium strategies, it is better to not reveal anything so as to maintain the average (prior) belief of the buyer. ■

Example C. (The Winner's Curse) The winner's curse, the phenomenon whereby bidders in a common value auction fail to condition on having received the most optimistic private signal and hence overbid, can be captured in a very similar fashion to Example 1. To see this, consider a stylized version of the problem wherein the object for auction has a common value, taking values on some set V. Each bidder $i = 1 \dots n$ receives a private signal $s \in S$ correlated with the value of the object.

After observing s how should a rational bidder bid? She should calculate her value conditional on s *and* the fact that she wins the auction, since her bid is irrelevant any other time. We can model this, like the previous example, with a state-space that is the product

space of three sources of uncertainty $\Omega = V \times S \times V$: for $(v, s, u) \in \Omega$, v is the value of the object, s the private signal of the bidder and u the upper-bound of all other bidders' bids. The probability μ depends on the distribution of value, the signal generation process, and the strategic model in the bidder's head (which in an equilibrium would be the other bidders' true behavior). Hence the optimal bid, b, must be based on maintaining a value $\mathbb{E}(v \mid V \times \{s\} \times \{u \mid u < b\})$.

To capture the winner's curse, we must represent the bidder's flawed hypothetical reasoning. In particular, her propensity to ignore the fact that winning the auction is informative about other bidders' private information and hence the value of the object itself. As above, ignoring this fact is tantamount to ignoring a dimension of the state-space: define π over Ω via

$$\pi(v, s, u) = \bigcup_{u' \in V} (v, s, u')$$

for $(v, s) \in V \times S$, and extend π to 2^Ω via $\pi(E) = \bigcup_{\omega \in E} \pi(\omega)$. Now, the bidder, after seeing signal s maintains an expected value for the object $\mathbb{E}(v \mid \pi(V \times \{s\} \times \{u \mid u < b\})) = \mathbb{E}(v \mid V \times \{s\} \times V)$ which is independent of the other bidders' strategies. ■

Example D. (The Monty Hall Paradox) At the end of the game show *Let's Make a Deal*, the contestant is faced with three doors. Behind two of the doors sits a goat and behind the third, a prize. The contestant is asked to choose a door, after which the host of the show, Monty Hall, opens one of the unchosen doors and reveals a goat. Crucially, Monty himself knows the contents of the doors and *always* opens a door with a goat behind it.

The contestant is then afforded a final choice: stick with her chosen door or switch to the other (closed) door. Assuming all randomizations (i.e., the allocation of the prize and Monty's revelation) were uniform,[3] what should the contestant do? We can analyze the uncertainty faced by the contestant using a simple 4-state model.

By possibly relabeling the doors, we can assume without loss of generality that the contestant initially selects door 1. Then the 4 relevant states are: $\Omega = \{\omega_{12}, \omega_{13}, \omega_{23}, \omega_{32}\}$, where ω_{ij} represents the state where the prize is hidden behind door i and Monty opens door j, see Figure 2. Our assumption of uniform randomization indicates the probability over the four states (in the same order as listed above) is $\mu = [\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3}]$.

Let P_i denote the event that the prize is behind door i and O_j the event that Monty opens door j. Assuming the prize provides 1 util and losing 0, the (objective) value of betting on

[3] And, perhaps more controversially, assuming that the contestant prefers the prize to a lovely pet goat.

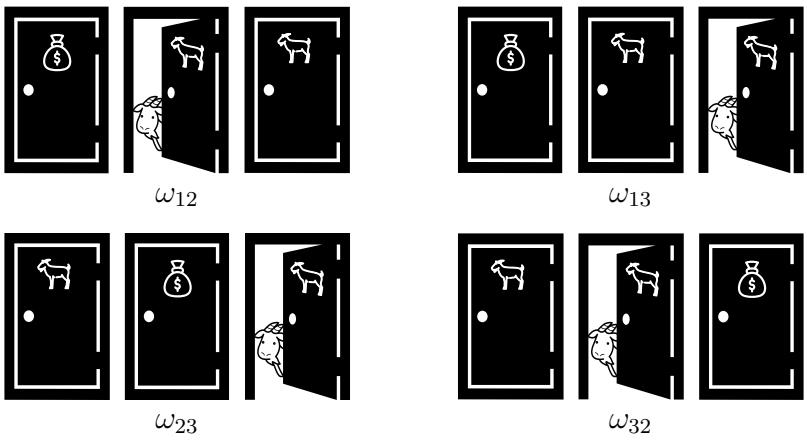


Figure 2: States in the Monty Hall Problem. Of course, the symbols on the doors indicating their contents are for our benefit and are invisible to the contestant when making her decision.

door 1 after Monty opens door 2 is

$$V(1_{P_1} \mid O_2) = \mu(P_1 \mid O_2) = \tfrac{\mu(P_1 \cap O_2)}{\mu(O_2)} = \tfrac{\mu(\{\omega_{12}\})}{\mu(\{\omega_{12},\omega_{32}\})} = \tfrac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3},$$

and the value of betting on door 3 is

$$V(1_{P_3} \mid O_2) = \mu(P_3 \mid O_2) = \tfrac{\mu(P_3 \cap O_2)}{\mu(O_2)} = \tfrac{\mu(\{\omega_{32}\})}{\mu(\{\omega_{12},\omega_{32}\})} = \tfrac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

The same logic applies when Monty opens door 3 and so in every case, the contestant should (objectively) strictly prefer to switch doors.

Of course, contestants on the actual *Let's Make a Deal* as well as countless subsequent thought experimenters have resisted this simple conclusion, even when presented with an analysis like the one above. Their flawed logic often runs as follows: There is no initial bias between door 1 and door 3 as evidenced by $\mu(P_1) = \mu(P_3) = \frac{1}{3}$, and Monty's revelation of a goat behind door 2 does not directly concern the contents of doors 1 and 3, and so, even once door 2 has been eliminated the probability of winning is independent of the choice to switch.

This logic, although tempting, is specious. Its failure is precisely a failure of hypothetical reasoning. Consider the following two hypotheses:

$O_2 =$ "Monty opens door 2", and

$\Omega \setminus P_2 =$ "The prize is not behind door two."

The contestant who does not strictly prefer to switch does so because she conflates these two hypotheses, mistakenly perceiving $(\Omega \setminus P_2) \Rightarrow O_2$. While plausible, this is incorrect. Such a DM fails to incorporate the additional information contained in the hypothesis O_2—namely, that Monty could have chosen door 3 if the prize is behind door 1, but this did not happen—and so, fails to eliminate ω_{13}.

This is captured by the interpretation $\pi : 2^\Omega \to 2^\Omega$ which maps a hypothesis G to $G \cup \{\omega_{13}\}$ whenever $O_2 \subseteq G$ and $G \cup \{\omega_{12}\}$ whenever $O_3 \subseteq G$, and is the identity elsewhere. In other words, while the hypothesis O_2 objectively corresponds to the event $\{\omega_{12}, \omega_{32}\}$, the contestant misinterprets it as $\{\omega_{12}, \omega_{13}, \omega_{32}\}$. It is easy to check that π is coherent.

Using the same probabilistic beliefs but under this new interpretation:

$$V_\pi(1_{P_1} \mid O_2) = \mu(\pi(P_1) \mid \pi(O_2)) = \tfrac{\mu(\{\omega_{12},\omega_{13}\})}{\mu(\{\omega_{12},\omega_{13},\omega_{32}\})} = \frac{\frac{2}{6}}{\frac{2}{3}} = \frac{1}{2}, \text{ and}$$
$$V_\pi(1_{P_3} \mid O_2) = \mu(\pi(P_3) \mid \pi(O_2)) = \tfrac{\mu(\{\omega_{32}\})}{\mu(\{\omega_{12},\omega_{13},\omega_{32}\})} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2},$$

the HEU contestant indeed sees no benefit to switching doors.

This model can be tested directly via the contestant's value for bets over hypotheses: consider the bets

1_{O_2} that pays 1 if O_2 is true and 0 otherwise, and

$1_{O_2 \cup (\Omega \setminus P_2)}$ that pays 1 if either O_2 or $\Omega \setminus P_2$ is true and 0 otherwise.

A contestant who mistakenly conflates the hypotheses will value these acts equally, as she does not see any added benefit to winning a bet after $(\Omega \setminus P_2)$, should she already win after O_2. Therefore, the contestant's preference over these two bets *directly reveals* her understanding of the relevant hypotheses, and provides a specific and observable test of the model. ■

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